Layer 9: Surface Enzyme Kinetics

Cleavage Kinetics of Surface-Immobilised SGKIPRTLTA by MMP-9

Eton iGEM 2026, Dry Lab | Layer 9 of 17-Layer Biosensor Pipeline | `surface_kinetics.py`

I. Physical Picture

The biosensor probe SGKIPRTLTA is surface-immobilised on the gold electrode via a Cys-thiol anchor and carries a methylene blue (MB) redox label at the C-terminal lysine. At surface concentration Γ_0 (mol cm^{-2}), the intact probe keeps MB tethered close to the electrode; MMP-9 cleavage releases the MB-bearing fragment into solution, decreasing the SWV peak current. The fractional cleavage $F_k(t)$ is therefore the central output of this layer and the direct precursor of the ratiometric signal $\Delta R/R_0$ (Layer 13).

The governing kinetics are heterogeneous: the enzyme (E, MMP-9) diffuses freely in solution while the substrate (S, probe) is fixed at the surface. Classical solution-phase Michaelis–Menten equations, derived under the condition $[S] \gg [E]$, do not apply in this configuration because the total molar quantity of surface-immobilised peptide ($\Gamma_0 \cdot A \sim 10$ fmol) can be comparable to or less than the bulk enzyme content. Anne and Demaille [1] derived exact kinetic equations for this heterogeneous surface mechanism without invoking the steady-state approximation.

The enzyme input $[E]_t = [\text{MMP-9}]_{\text{free}}$ is taken directly from the Layer 0 wound biology ODE output: 30 nM (chronic wound, 30-min equilibration) and 6 nM (healing wound). TIMP-1 inhibition is fully accounted for in these values; no further correction is needed here.

II. Three-State Surface ODE System

Reaction mechanism

The surface enzyme reaction follows the Michaelis–Menten mechanism with the substrate anchored to the surface [1]:

$$S + E \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_{\text{cat}}} P + E \tag{II.1}$$

where S, ES, P are surface species with concentrations Γ_S, Γ_{ES}, Γ_P (mol cm^{-2}); k_1 (M^{-1}s^{-1}) is the second-order binding rate constant; k_{-1} (s^{-1}) is the dissociation rate constant; and k_{cat} (s^{-1}) is the catalytic rate constant. The enzyme concentration at the electrode surface $[E]_t$ (M) is treated as constant throughout (justified in Section V).

Governing ODEs

The time evolution of each surface species is [1] (Eq. 1):

$$\frac{d\Gamma_S}{dt} = k_{-1}\,\Gamma_{ES} - k_1[E]_t\,\Gamma_S, \qquad \frac{d\Gamma_{ES}}{dt} = k_1[E]_t\,\Gamma_S - (k_{-1} + k_{\text{cat}})\,\Gamma_{ES}, \qquad \frac{d\Gamma_P}{dt} = k_{\text{cat}}\,\Gamma_{ES} \tag{II.2}$$

with surface conservation and initial conditions:

$$\Gamma_S + \Gamma_{ES} + \Gamma_P = \Gamma_0, \qquad \Gamma_S(0) = \Gamma_0, \quad \Gamma_{ES}(0) = \Gamma_P(0) = 0 \tag{II.3}$$

Introducing the unconverted substrate $\Gamma^* = \Gamma_S + \Gamma_{ES}$ and defining $k_1^* = k_1[E]_t$, the system reduces to the two-variable form used by Anne and Demaille [1] for the Laplace analysis:

$$\frac{d\Gamma_{ES}}{dt} = k_1^*\,\Gamma^* - (k_1^* + k_{-1} + k_{\text{cat}})\,\Gamma_{ES}, \qquad \frac{d\Gamma^*}{dt} = -k_{\text{cat}}\,\Gamma_{ES} \tag{II.4}$$

III. Analytic Solutions

Steady-state approximation

Setting $d\Gamma_{ES}/dt = 0$ in Eq. (II.4) and substituting the conservation constraint yields the steady-state progress curve [1] (Eq. 3):

$$\theta_P^{\text{ss}} = \frac{\Gamma_P}{\Gamma_0} = 1 - \exp(-k_{\text{eff}}\,t), \qquad k_{\text{eff}} = \frac{k_{\text{cat}}}{1 + K_M/[E]_t}, \quad K_M = \frac{k_{-1} + k_{\text{cat}}}{k_1} \tag{III.1}$$

This is the Langmuirian (LG) progress curve. In the linear regime ($[E]_t \ll K_M$): $k_{\text{eff}} \approx (k_{\text{cat}}/K_M)[E]_t \equiv k_{\text{eff}}^{\text{surf}}[E]_t$, giving a simple pseudo-first-order decay.

Exact biexponential solution

The reduced system Eq. (II.4) is linear and solved exactly via Laplace transform [1]. Define the composite parameters:

$$k_1^* = k_1[E]_t, \qquad \sigma = k_1^* + k_{-1} + k_{\text{cat}}, \qquad \Delta = \sigma^2 - 4k_1^* k_{\text{cat}} > 0 \tag{III.2}$$

The exact solutions are [1] (Eqs. 4–5):

$$\Gamma_{ES}(t) = \frac{k_1^* \Gamma_0}{\sqrt{\Delta}} \left[\exp\left(-\frac{\sigma - \sqrt{\Delta}}{2}\, t \right) - \exp\left(-\frac{\sigma + \sqrt{\Delta}}{2}\, t \right) \right] \tag{III.3}$$

$$\Gamma^*(t) = \frac{\Gamma_0}{2\sqrt{\Delta}} \left[(\sigma + \sqrt{\Delta}) \exp\left(-\frac{\sigma - \sqrt{\Delta}}{2}\, t \right) - (\sigma - \sqrt{\Delta}) \exp\left(-\frac{\sigma + \sqrt{\Delta}}{2}\, t \right) \right] \tag{III.4}$$

The fractional product coverage (fractional cleavage) is then:

$$F_k(t) = \theta_P(t) = 1 - \frac{\Gamma^*(t)}{\Gamma_0} \tag{III.5}$$

λ parameter and dimensionless universal form

Anne and Demaille [1] introduce the dimensionless kinetic zone parameter:

$$\lambda = \frac{\sigma}{\sqrt{\Delta}} = \frac{1}{\sqrt{1 - 4k_1^* k_{\text{cat}}/\sigma^2}} = \frac{1}{\sqrt{1 - 4k_1 k_{\text{cat}}[E]_t \,/\, (k_1^* + k_{-1} + k_{\text{cat}})^2}} \geq 1 \tag{III.6}$$

and the dimensionless time $\tau = \sigma(1 - 1/\lambda)\, t$. In these variables the exact progress curve Eq. (III.5) becomes [1] (Eq. 7):

$$\theta_P(\tau, \lambda) = 1 - \frac{1}{2} \left\{ (1 + \lambda) \exp\left[-\frac{\tau}{2} \right] + (1 - \lambda) \exp\left[-\frac{1}{2} \frac{\lambda + 1}{\lambda - 1}\, \tau \right] \right\} \tag{III.7}$$

This single equation captures the entire range of possible kinetic behaviours through the value of λ alone [1]. Two limiting cases are:

- **LG limit** ($\lambda \to 1$): Eq. (III.7) reduces to $\theta_P = 1 - \exp(-\tau/2)$, a simple monoexponential. The steady-state approximation is exact. Diagnostic: $t_{1/2}/t_{1/4} \to 2.41$ [1].
- **FS limit** ($\lambda \gg 1$): Eq. (III.7) reduces to [1] (Eq. 8):

$$\theta_P = 1 - \left(1 + \frac{\tau}{2} \right) \exp\left(-\frac{\tau}{2} \right) \tag{III.8}$$

  The progress curve is S-shaped (sigmoidal foot at short times). The steady-state approximation overestimates θ_P and fails here. Diagnostic: $t_{1/2}/t_{1/4} \to 1.75$ [1].

The zone diagram [1] (Fig. 4) maps these regimes in $\log(k_{-1}/k_{\text{cat}}) - \log([E]_t/K_M)$ space. The LG zone applies when $k_{-1}/k_{\text{cat}} \gg \Phi$, where $\Phi = 4[E]_t/K_M \,/(1 + [E]_t/K_M)^2$, or unconditionally when $[E]_t/K_M \leq 0.2$ or $[E]_t/K_M \geq 5$.

IV. Kinetic Regime at Clinical MMP-9 Concentrations

$\lambda \approx 1$: the Langmuirian limit is guaranteed

At clinical concentrations $[E]_t \in [0.19, 2.17]$ nM and any physically realisable binding rate constant $k_1 \leq 10^8$ $\text{M}^{-1}\text{s}^{-1}$ (diffusion limit for a 92 kDa enzyme):

$$k_1^* = k_1[E]_t \leq 10^8 \times 2.17 \times 10^{-9} = 0.22 \text{ s}^{-1} \tag{IV.1}$$

The solution-phase value $k_{\text{cat}}^{\text{sol}} = 86.4$ s^{-1} [2] is an upper bound for the surface k_{cat}. In the regime $k_1^* \ll k_{\text{cat}}$:

$$\Delta \approx (k_{-1} + k_{\text{cat}})^2 + 2k_1^*(k_{-1} - k_{\text{cat}}) \approx (k_{-1} + k_{\text{cat}})^2, \quad \lambda = \frac{k_1^* + k_{-1} + k_{\text{cat}}}{\sqrt{\Delta}} \approx 1 + O\left(\frac{k_1^*}{k_{\text{cat}}} \right) \tag{IV.2}$$

Since $k_1^*/k_{\text{cat}}^{\text{sol}} \leq 2.5 \times 10^{-3}$, we have $\lambda - 1 < 3 \times 10^{-3}$ regardless of the unknown surface k_{-1}. The system is firmly in the LG zone.

Collapse to monoexponential

In the LG limit, the exact biexponential Eq. (III.7) degenerates to the Langmuirian monoexponential irrespective of the k_{-1}/k_{cat} ratio:

$$\boxed{F_k(t) = 1 - \exp\!\left(-k_{\text{eff}}^{\text{surf}}\,[E]_t\,t\right)} \tag{IV.3}$$

where $k_{\text{eff}}^{\text{surf}} \equiv k_{\text{cat}}^{\text{surf}}/K_M^{\text{surf}}$ is the single measurable surface rate constant. This result is notable: the unknowns k_1, k_{-1}, and $k_{\text{cat}}^{\text{surf}}$ individually are inaccessible at clinical $[E]_t$, but the product $k_{\text{eff}}^{\text{surf}}[E]_t$ is directly fit from the progress curve. The complex accumulates at sub-ppm levels ($\Gamma_{ES}/\Gamma_0 < 10^{-3}$) and is kinetically invisible.

The solution upper bound is $k_{\text{eff}}^{\text{surf}} \le k_{\text{cat}}^{\text{sol}}/K_m^{\text{sol}} = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ [2]. Surface immobilisation reduces this by 2–100× depending on steric accessibility (Layer 8); the surface value must be measured on the fabricated electrode.

V. Dimensionless Analysis

Three dimensionless numbers characterise the device operating regime and validate the sequential layer architecture.

Kinetic zone parameter λ

From Section IV: $\lambda \approx 1$ for all clinical $[E]_t$. The monoexponential Eq. (IV.3) is the appropriate model. The zone parameter provides an experimental diagnostic: if the measured progress curve shows $t_{1/2}/t_{1/4}$ significantly below 2.41, the system has left the LG zone and the full exact solution Eq. (III.7) must be used.

Damköhler number: transport does not limit

The Damköhler number compares the surface reaction rate to the enzyme delivery rate by diffusion [4]:

$$\text{Da} = \frac{k_{\text{eff}}^{\text{surf}}\,[E]_t\,h}{D_E} \tag{V.1}$$

where $h \approx 100\ \mu$m is the diffusion layer height (Layer 3) and $D_E \approx 4.7 \times 10^{-7}\ \text{cm}^2\text{s}^{-1}$ is the MMP-9 diffusion coefficient (Stokes–Einstein, Layer 3). At the solution upper bound with $[E]_t = 2.17$ nM:

$$\text{Da}^{\max} = \frac{67{,}500 \times 2.17 \times 10^{-9} \times 10^{-2}}{4.7 \times 10^{-7}} \approx 3.1 \times 10^{-3} \ll 1 \tag{V.2}$$

The electrode surface sees the bulk enzyme concentration to within $< 0.4\%$ at all times. For any plausible surface $k_{\text{eff}}^{\text{surf}} < 67{,}500\ \text{M}^{-1}\text{s}^{-1}$, Da is smaller still. The Layer 3 mass-transport correction is negligible; $[E]_t$ in Eq. (IV.3) is the bulk concentration from Layer 0 throughout.

The Damköhler criterion is consistent with the general analysis of Squires et al. [4]: for $\text{Da} \ll 1$ the sensor operates in the reaction-limited regime where signal is proportional to the surface reaction rate, not the diffusion flux, and the response is linear in enzyme concentration at short times.

Enzyme conservation: $[E]_t$ is constant

MMP-9 is a true catalyst: it is released intact after each cleavage event (Eq. (II.1)). The bulk enzyme concentration $[E]_t$ therefore does not decrease as Γ_P increases. This is distinct from, for example, irreversible inhibitors. The constancy of $[E]_t$ is a consequence of the catalytic mechanism, not an approximation, and holds for all t regardless of F_k.

VI. Coupled PDE–ODE Numerical Treatment

Full coupled system

For completeness, and to verify the $\text{Da} \ll 1$ approximation, the full coupled system – Layer 3 diffusion PDE and Layer 9 surface ODE – is solved simultaneously:

$$\frac{\partial [E]}{\partial t} = D_E\,\frac{\partial^2 [E]}{\partial z^2}, \quad 0 < z < h \tag{VI.1}$$

$$\frac{d\Gamma_{ES}}{dt} = k_1^*\Gamma^* - (k_1^* + k_{-1} + k_{\text{cat}})\Gamma_{ES} \quad \text{with } k_1^* = k_1[E](z{=}0,t) \tag{VI.2}$$

with boundary conditions:

$$D_E\,\frac{\partial [E]}{\partial z}\bigg|_{z=0} = k_{\text{cat}}\,\Gamma_{ES}(t) \quad \text{(flux = cleavage consumption)}, \qquad [E](h,t) = [E]_{\text{bulk}} \tag{VI.3}$$

Solved numerically: Crank–Nicolson finite difference for the PDE, RK4 for the surface ODE, coupled at each timestep through the $z = 0$ boundary condition.

Verification of the decoupled limit

For Da $\ll 1$, the PDE solution gives $[E](0,t) \approx [E]_{\text{bulk}}$ to within $\text{Da}/(1+\text{Da}) < 0.4\%$. The surface ODE then has a constant-coefficient enzyme input and reduces exactly to Eq. (IV.3). The coupled simulation (`surface_kinetics.py`) confirms this: the full PDE–ODE $F_k(t)$ is indistinguishable from the analytic monoexponential at all clinical $[E]_t$ values. The decoupled monoexponential is therefore used for all subsequent analysis.

VII. $F_k(t)$ Output and Optimal Readout Time

Progress curves

From Eq. (IV.3), the fractional cleavage at incubation time T:

$$F_k(T) = 1 - \exp\!\left(-k_{\text{eff}}^{\text{surf}}\,[E]_t\,T\right) \tag{VII.1}$$

In the linear regime ($k_{\text{eff}}^{\text{surf}}[E]_t T \ll 1$):

$$F_k(T) \approx k_{\text{eff}}^{\text{surf}}\,[E]_t\,T \tag{VII.2}$$

The ratio of chronic to healing fractional cleavage is:

$$\frac{F_k^{\text{chronic}}}{F_k^{\text{healing}}} \approx \frac{[E]_{\text{chronic}}}{[E]_{\text{healing}}} = \frac{30\text{ nM}}{6\text{ nM}} = 5\times \tag{VII.3}$$

independent of $k_{\text{eff}}^{\text{surf}}$ and T. The $5\times$ discrimination ratio is therefore robust across the entire plausible range of surface rate constants.

Optimal readout time T^*

In the linear regime, $F_k \propto [E]_t T$, so the ratio F_k^c/F_k^h is always $5\times$ regardless of T. The optimal incubation time is then as long as possible before signal saturation ($F_k \to 1$).

In the exponential (nonlinear) regime, the *absolute* signal difference $\Delta F_k = F_k^{\text{chronic}} - F_k^{\text{healing}}$ varies with T. Maximising ΔF_k gives the optimal readout time. Setting $d(\Delta F_k)/dT = 0$:

$$\begin{aligned} k_{\text{eff}}^{\text{surf}}\,[E]_c\,\exp\!\left(-k_{\text{eff}}^{\text{surf}}[E]_c T\right) &= k_{\text{eff}}^{\text{surf}}\,[E]_h\,\exp\!\left(-k_{\text{eff}}^{\text{surf}}[E]_h T\right) \\ \Rightarrow \quad T^* &= \frac{\ln([E]_c/[E]_h)}{k_{\text{eff}}^{\text{surf}}\,([E]_c - [E]_h)} \end{aligned} \tag{VII.4}$$

With $[E]_c = 30$ nM, $[E]_h = 6$ nM, and ratio $r = 5$:

$$T^* = \frac{\ln 5}{k_{\text{eff}}^{\text{surf}} \times 24\text{ nM}} = \frac{1.609}{k_{\text{eff}}^{\text{surf}} \times 2.4 \times 10^{-8}\text{ M}} \tag{VII.5}$$

$k_{\text{eff}}^{\text{surf}}$ ($\text{M}^{-1}\text{s}^{-1}$)	T^* (s)	T^* (min)
2,000	33,500	558
10,000	6,700	112
30,000	2,230	37
67,500 (sol. UB)	990	16

The target incubation window of 30–60 min corresponds to $k_{\text{eff}}^{\text{surf}} \approx 13{,}000$–$27{,}000$ $\text{M}^{-1}\text{s}^{-1}$. If the measured $k_{\text{eff}}^{\text{surf}}$ falls significantly below this, the incubation time must be extended accordingly.

VIII. Temperature Dependence

Arrhenius correction

The surface rate constant follows Arrhenius temperature dependence:

$$k_{\text{eff}}^{\text{surf}}(T_{\text{amb}}) = k_{\text{eff}}^{\text{surf}}(T_{\text{ref}})\,\exp\!\left[-\frac{E_a}{R}\left(\frac{1}{T_{\text{amb}}} - \frac{1}{T_{\text{ref}}}\right)\right] \tag{VIII.1}$$

where E_a is the activation energy and $T_{\text{ref}} = 310$ K (37°C). The relative sensitivity to ambient temperature is:

$$\frac{1}{k_{\text{eff}}}\frac{dk_{\text{eff}}}{dT_{\text{amb}}} = \frac{E_a}{R\,T_{\text{amb}}^2} \tag{VIII.2}$$

The activation energy E_a for MMP-9 surface cleavage of SGKIPRTLTA is not confirmed from any paper in `papers/layer_09/`. Using a typical metalloproteinase value $E_a \approx 45$ kJ mol^{-1} (unconfirmed placeholder):

$$\left.\frac{d\ln k_{\text{eff}}}{dT}\right|_{310\,\text{K}} \approx \frac{45{,}000}{8.314 \times 310^2} \approx 5.6\%\text{ per °C} \tag{VIII.3}$$

Propagation to F_k

Let $x = k_{\text{eff}}^{\text{surf}}[E]_t T$. The temperature sensitivity of F_k is:

$$\frac{\partial F_k}{\partial T_{\text{amb}}} = (1 - F_k)\, x\, \frac{E_a}{R T_{\text{amb}}^2} = (1 - F_k)\, |\ln(1 - F_k)|\, \frac{E_a}{R T_{\text{amb}}^2} \tag{VIII.4}$$

At $F_k = 0.1$ (early signal, $x = 0.105$) and $\pm 1°\text{C}$ uncertainty: $\delta F_k \approx 0.1 \times 0.105 \times 0.056 \approx 6 \times 10^{-4}$, a 0.6% absolute error on F_k. At $F_k = 0.5$ ($x = 0.693$): $\delta F_k \approx 0.5 \times 0.693 \times 0.056 \approx 1.9\%$.

Temperature control or measurement is required if absolute F_k accuracy $< 2\%$ is needed. The ratiometric signal $\Delta R/R_0$ is similarly affected since F_k enters linearly (Layer 13). The discrimination *ratio* F_k^c/F_k^h is less sensitive: at short times both F_k^c and F_k^h scale with the same $k_{\text{eff}}(T)$, so the ratio $[E]_c/[E]_h = 5\times$ is temperature-invariant.

IX. Multi-Protease Competition

General competition model

Wound fluid contains multiple proteases that may cleave SGKIPRTLTA. In the first-order (linear) regime, contributions add independently:

$$\frac{dF_k}{dt} = k_{\text{obs,total}} = \sum_j k_{\text{eff},j}^{\text{surf}}\, [E_j]_t \tag{IX.1}$$

The fractional contribution of interferent j to the observed cleavage signal is:

$$\xi_j = \frac{k_{\text{eff},j}^{\text{surf}}\, [E_j]_t}{k_{\text{obs,total}}} \tag{IX.2}$$

MMP-9 selectivity requires $\xi_{\text{MMP-9}} \approx 1$, i.e. $k_{\text{eff,MMP-9}}^{\text{surf}}[E_{\text{MMP-9}}] \gg \sum_{j \neq \text{MMP-9}} k_{\text{eff},j}^{\text{surf}}[E_j]$.

Known interferents and status

Interferent	$[E_j]$ source	$k_{\text{eff},j}^{\text{surf}}$ status	Risk
HNE	30 nM chronic (Layer 0, Trengove 1999)	Unconfirmed	Moderate
MMP-2	Wound fluid present, conc. unconfirmed	Unconfirmed	Low–moderate
MMP-8	Wound fluid present, conc. unconfirmed	Unconfirmed	Low

No paper in `papers/layer_09/` *provides k_{eff}^{surf} for any interferent protease on SGKIPRTLTA. The competition fractions ξ_j cannot be computed until these are measured.* HNE is the highest-priority interferent given its confirmed elevated concentration in chronic VLU and its Ser-protease activity overlapping with some MMP substrates.

X. Uncertainty Propagation

Analytic uncertainty in F_k

Let $x = k_{\text{eff}}^{\text{surf}}[E]_t T$, so $F_k = 1 - e^{-x}$. The partial derivatives are:

$$\frac{\partial F_k}{\partial k_{\text{eff}}} = \frac{x}{k_{\text{eff}}}\, e^{-x}, \qquad \frac{\partial F_k}{\partial [E]_t} = \frac{x}{[E]_t}\, e^{-x}, \qquad \frac{\partial F_k}{\partial T} = \frac{x}{T}\, e^{-x} \tag{X.1}$$

By standard propagation:

$$\sigma_{F_k} = e^{-x}\, x \sqrt{\left(\frac{\sigma_{k_{\text{eff}}}}{k_{\text{eff}}}\right)^2 + \left(\frac{\sigma_{[E]}}{[E]_t}\right)^2 + \left(\frac{\sigma_T}{T}\right)^2} = (1 - F_k)\, |\ln(1 - F_k)|\, \sigma_{\text{rel,total}} \tag{X.2}$$

Dominant uncertainty sources

Source	Relative uncertainty σ/μ	Status
$k_{\text{eff}}^{\text{surf}}$	Unknown; plausible range 2,000–67,500 $\text{M}^{-1}\text{s}^{-1}$ (factor 34×)	Dominant until measured
$[E]_t$	$\sim$ 15–20% (Layer 0 ODE; healing estimate approximate)	Secondary
T (incubation)	$\sim$ 1–2% (timer precision)	Negligible
T_{amb}	$\sim 0.6\%$ per °C (Eq. VIII.3)	Tertiary

The dominant uncertainty is $k_{\text{eff}}^{\text{surf}}$, which spans more than an order of magnitude before electrode calibration. All F_k predictions are therefore presented as parametric sweeps over the plausible $k_{\text{eff}}^{\text{surf}}$ range. Once the fabricated electrode is measured (Layer 15 calibration), $k_{\text{eff}}^{\text{surf}}$ is pinned to ±5–10% and σ_{F_k} is determined by $\sigma_{[E]}$ alone.

XI. Parameter Table and Layer Outputs

Parameter	Description	Value	Source	Confirmed
$k_{\text{cat}}^{\text{sol}}$	Solution-phase catalytic rate for SGKIPRTLTA	86.4 s^{-1}	Kridel 2001 [2] Table II	Yes
K_m^{sol}	Solution-phase Michaelis constant	1.28 mM	Kridel 2001 [2] Table II	Yes
$(k_{\text{cat}}/K_m)^{\text{sol}}$	Solution efficiency (UB for surface)	67,500 $\text{M}^{-1}\text{s}^{-1}$	Kridel 2001 [2] Table II	Yes
$[E]_t^{\text{chronic}}$	Free MMP-9 at 30-min read, chronic	30 nM	Layer 0 ODE	Yes
$[E]_t^{\text{healing}}$	Free MMP-9 at 30-min read, healing	6 nM	Layer 0 ODE	Yes
Γ_0	Initial surface probe density	$\sim$5 pmol cm^{-2}	Layer 7 (device-planned)	No
$k_{\text{eff}}^{\text{surf}}$	Surface rate constant	To be measured	Layer 15 calibration	No
K_m^{surf}	Surface Michaelis constant	To be measured	Layer 15 calibration	No
E_a	Activation energy for surface cleavage	$\sim$45 kJ mol^{-1} placeholder	Not in `papers/layer_09/`	No
A	Electrode area	$\sim$0.002 cm^2	Device-planned	No
$k_{\text{eff},j}^{\text{surf}}$	Surface rate constant for interferents	Unknown	Not in `papers/layer_09/`	No

Layer outputs.

- $F_k(T)$: fractional cleavage as function of incubation time → Layer 13 (ratiometric signal $\Delta R/R_0$)
- $k_{\text{obs}} = k_{\text{eff}}^{\text{surf}}[E]_t$: pseudo-first-order rate → Layer 14 (noise budget)
- T^*: optimal incubation time (Eq. (VII.4)) → device protocol
- $k_{\text{eff}}^{\text{surf}}$ (post-calibration) → Layer 1 (dilution factor), Layer 15 (calibration curve)

XII. Limitations

1. **$k_{\text{eff}}^{\text{surf}}$ is unknown; all F_k predictions are parametric.** The surface rate constant $k_{\text{eff}}^{\text{surf}}$ must be measured on the fabricated electrode before any absolute signal or LOD can be specified. The solution upper bound $k_{\text{cat}}^{\text{sol}}/K_m^{\text{sol}} = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ [2] is the starting assumption; surface immobilisation typically reduces this by 2–100× (Layer 8 steric correction, Adjémian 2010 [3]). Until calibration, the plausible range spans 2,000–67,500 $\text{M}^{-1}\text{s}^{-1}$ and T^* spans 16 min to > 9 h.

2. **K_M^{surf} and the LG zone assumption require experimental verification.** The $\lambda \approx 1$ conclusion holds for any $k_1 \leq 10^8\ \text{M}^{-1}\text{s}^{-1}$ and $[E]_t \leq 2.17$ nM. However, if the effective surface K_M^{surf} is pushed above ~ 100 nM by steric crowding (putting $[E]_t/K_M^{\text{surf}}$ above 0.02), the system could enter a regime where the exact biexponential Eq. (III.7) is needed for accurate fitting. The diagnostic ratio $t_{1/2}/t_{1/4} \to 2.41$ (LG) vs. $\to 1.75$ (FS) should be measured [1].

3. **Temperature dependence is unconfirmed.** The Arrhenius activation energy E_a for MMP-9 cleavage of SGKIPRTLTA on a surface is not reported in any paper in `papers/layer_09/`. The 5.6% per °C figure is a placeholder based on typical metalloproteinase values. A temperature-controlled incubation or in-situ temperature measurement is required for absolute F_k accuracy.

4. **Interferent cross-reactivity is unquantified.** The surface rate constants $k_{\text{eff},j}^{\text{surf}}$ for HNE, MMP-2, MMP-8, and other wound-fluid proteases on SGKIPRTLTA are not reported in any paper in `papers/layer_09/`. The selectivity fractions ξ_j cannot be computed. HNE is the priority interferent given its confirmed elevated chronic wound concentration (30 nM, Layer 0 [2]) and potential serine-protease activity on the substrate.

5. **Solution-to-surface K_m conversion requires δ_{eff} from Layer 8.** The effective surface Michaelis constant is related to the solution value by $K_M^{\text{surf}} \approx K_m^{\text{sol}}\,\delta_{\text{eff}}$, where δ_{eff} is the effective thickness of the accessible probe layer (Layer 8, unconfirmed). Until δ_{eff} is established, K_M^{surf} cannot be predicted from K_m^{sol} alone.

References

All papers in `electrochemical_device/papers/layer_09/`.

[1] Anne A, Demaille C (2012). Kinetics of enzyme action on surface-attached substrates: a practical guide to progress curve analysis in any kinetic situation. *Langmuir* 28:14665–14671. **anne_demaille2012.pdf** – ODE system (Eq. 1), exact biexponential solution (Eqs. 4–5), λ parameter (Eq. 6 region), universal progress curve (Eq. 7), FS limit (Eq. 8), zone diagram (Fig. 4): all confirmed from text.

[2] Kridel SJ, Chen E, Howard EW, Bhatt L, Strongin AY (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276:20572–20578. **kridel2001.pdf** – $k_{\text{cat}} = 86.4\ \text{s}^{-1}$, $K_m = 1.28$ mM, $k_{\text{cat}}/K_m = 67{,}500\ \text{M}^{-1}\text{s}^{-1}$ for SGKIPRTLTA (peptide A11), Table II confirmed.

[3] Adjémian J, Anne A, Cauet G, Demaille C (2010). Cleavage-sensing peptide-enzyme multilayer gold surfaces for the parallel, selective, and sensitive assay of trypsin and plasmin activities by electrochemical methods. *Langmuir* 26:10347–10356. **adjemian2010.pdf** – experimental surface cleavage yields and Γ_0 context.

[4] Squires TM, Messinger RJ, Manalis SR (2008). Making it stick: convection, reaction and diffusion in surface-based biosensors. *Nat Biotechnol* 26:417–426. **squires2008.pdf** – Damköhler number framework for reaction-limited vs. transport-limited biosensor regimes.